

# Coupling-gated cortical coherence: what a field theory of conscious unity must assume, and what it predicts

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## **Abstract**

A field account of cortical coordination holds that spatially distributed neural processes are coupled in part through the endogenous extracellular electromagnetic field. We ask what such an account must assume to reach a globally consistent, self-referential state, and what it predicts before it gets there. Its physical core is a noisy mean-field phase model whose coupling drifts through a synchronization threshold. If recovery of coherence after sleep is gated by that crossing, it is not free to take an arbitrary time course. For as long as the phase distribution tracks the stationary branch, it must show a square-root onset and a von Mises concentration–coherence relation with no fitted shape parameter. The tracking condition is part of the prediction, and the simulations measure its cost: ramp speed sets the apparent onset exponent, so an exponent fitted to a driven recovery trace reports the drive as much as the mechanism. A coupling plasticity descending a squared-drift objective preserves coherence while moving away from the environmental structure it should learn, below a shuffled-label

comparison: dissipation minimization does not imply representational learning. A Lean 4 formalization composes finite capacity, Landauer’s bound, the noisy Kuramoto transition, sheaf gluing and Banach’s theorem into one conditional theorem, isolating the eight hypotheses it consumes; a common toy model satisfies all eight. We defend as a philosophical commitment, not a theorem, the identification of the glued section with the unity of a conscious episode and of its reflexive fixed point with a minimal self. Testing the account requires independent calibration of effective cortical coupling and an awakening experiment separating slow coupling change from alternative recovery mechanisms.

# 1 Introduction

Neural activity changes across sleep, awakening and anesthesia, but a change in synchrony does not by itself identify the physical interaction responsible for it. A field account makes a stronger proposal: spatially distributed neural processes are coordinated partly through the endogenous extracellular electromagnetic field, and changes in the medium can change that coordination (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Whether that mechanism is sufficient to support the collective state such accounts require remains an empirical question.

Such an account has three requirements, and they order what follows. Coupling must be strong enough to sustain coherent collective activity; the local descriptions that activity carries must be compatible enough to glue into a single one; and that glued state must support an accurate model of

15 itself. The first requirement is physical and testable with current methods,  
16 and most of this article is about testing it. The second and third are where  
17 the mathematics is, and where the assumptions accumulate.

18     The first requirement admits a definite claim. If cortical coupling is slowly  
19 gated—if recovery of coherence after sleep is driven by a coupling parameter  
20 drifting through a synchronization threshold—then the recovery is not free  
21 to take an arbitrary time course. For a recovery slow enough that the phase  
22 distribution tracks the stationary branch, the theory fixes a square-root onset  
23 and a one-parameter relation between phase concentration and coherence,  
24 with no fitted shape parameter. That is a constraint an experiment can  
25 meet or violate. How slow is slow enough is itself part of the claim, and  
26 our simulations bound it from the other side, by showing which parts of the  
27 constraint survive finite populations and finite ramp speeds and which do  
28 not.

29     The contribution has four parts. The first is the coupling-gated recovery  
30 hypothesis itself: a stationary coherent branch with square-root onset, a von  
31 Mises concentration–coherence relation, and an experimental protocol with  
32 failure conditions declared in advance (Sections 2 and 3). The second is a  
33 constraint on how such data may be analyzed. The apparent onset expo-  
34 nent measured from a finite ramp is a function of ramp speed across the  
35 sweep of Section 4, so an exponent fitted to a recovery trace measures the  
36 ramp at least as much as the mechanism—and that holds for any analysis  
37 reading a critical exponent off a driven transition, not only for ours. The

38 third is a negative result about plasticity: a coupling update descending  
 39 a squared-drift objective preserves coherence while moving *away* from the  
 40 environmental structure it is expected to learn, below a shuffled-label com-  
 41 parison. Dissipation minimization does not imply representational learning.  
 42 The fourth is a machine-checked composition (Section 5) in which finite in-  
 43 formation capacity, Landauer’s bound, a predictive dissipation bound, the  
 44 noisy Kuramoto transition, sheaf gluing and Banach’s fixed-point theorem  
 45 yield one conditional theorem, with the eight hypotheses that composition  
 46 consumes isolated and named; a common toy model satisfies all eight, so  
 47 their conjunction is not empty.

48 A prose table of assumptions would list the same eight items, so the fourth  
 49 contribution needs a word about method. What the Lean development adds  
 50 is that it makes the list non-negotiable. The characteristic failure in an  
 51 argument of this shape is not a false step but a silent change of subject:  
 52 a theorem about a stationary density read as a claim about a trajectory,  
 53 an inequality about a register read as a claim about learning, agreement of  
 54 phases read as compatibility of representations. Type-checking forbids each  
 55 of those substitutions, and an assumption needed to make a step go through  
 56 appears as a hypothesis in the term rather than as a word in a sentence. The  
 57 last of them is not merely forbidden but refuted by a constructed object: the  
 58 development carries a cover at a single common phase whose local densities  
 59 disagree where its patches meet, satisfying every other requirement of the  
 60 gluing theorem and failing compatibility alone (Section 5.2). That object

61 is what a prose table of assumptions cannot carry. Formalization does not  
62 make the assumptions true; it makes them impossible to lose.

63     The physical model and the formal composition belong in one article be-  
64 cause they are two halves of one commitment. The coupling-gated prediction  
65 is worth testing under this framework, rather than as one more synchroniza-  
66 tion model, only because the coherent state it produces is claimed to do  
67 further work: to support a globally consistent representation and a reflexive  
68 fixed point that we identify with phenomenal unity and a minimal self.

69     The distinction that does that work is between phases agreeing and de-  
70 scriptions fitting together, and it is concrete. Take two cortical regions whose  
71 receptive territories overlap, each carrying a probability distribution over  
72 what occupies the shared part. Their oscillators can be locked to the same  
73 phase at every instant while one region's distribution on the overlap places an  
74 object left of fixation and the other places it right. The phases agree; the de-  
75 scriptions do not, and no single distribution over the shared territory restricts  
76 to both. A theory that identifies unity with synchrony cannot mark that dif-  
77 ference, and so cannot separate a brain whose regions oscillate together from  
78 one whose regions are about the same world. The sheaf condition marks  
79 exactly it: local states glue to one global section when their restrictions to  
80 shared regions coincide, and otherwise not at all. Coherence enters as the  
81 physical condition under which compatibility can be maintained, not as a  
82 substitute for it, and the self-prediction map then asks the further ques-  
83 tion of whether the glued state contains an accurate model of itself. That

84 identification is a commitment, not a theorem, and Section 6 states it, ar-  
 85 gues for it, and says what would count against it. Its separation from the  
 86 physical tests is what makes both assessable: the cortical coupling hypothe-  
 87 sis and the recovery prediction can fail while the mathematical composition  
 88 stands, and the composition can stand while the identification is wrong. The  
 89 established ingredients (Landauer, 1961; Still et al., 2012; Sakaguchi, 1988;  
 90 Banach, 1922) therefore appear here for their biological composition, not for  
 91 any new fundamental physical law.

## 92 **2 A coupling-gated model of cortical coher-** 93 **ence**

### 94 **2.1 Dynamics and stationary solutions**

95 The finite stochastic model is

$$d\theta_i = \left[ \omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

96 Here  $D > 0$  is phase diffusion,  $W_i$  are independent Wiener processes, and  
 97  $K_{ij}$  has inverse-time units, with angles treated as dimensionless. The uni-  
 98 form mean-field convention is  $K_{ij} = K/N$ . Frequency heterogeneity and  
 99 phase diffusion are separate parameters; a frequency-spread width cannot be  
 100 substituted for  $D$  when applying the identical-frequency stationary theory.

101 For identical frequencies in their rotating frame, the assumed stationary  
 102 density is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}. \quad (2)$$

103 The  $I_n$  are modified Bessel functions, and  $\psi$  is the collective phase. Lean  
 104 verifies normalization, the density's circular order parameter, and the self-  
 105 consistency equation. At  $K \leq K_c = 2D$ , the only nonnegative self-consistent  
 106 order parameter is zero. At  $K > K_c$ , exactly one positive solution coex-  
 107 ists with zero; the positive branch is continuous at threshold and strictly  
 108 increasing with  $K$ .

109 The near-threshold expansion gives

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

110 This is a result about the stationary solution branch. Of the coherent branch  
 111 we prove existence, uniqueness, continuous emergence at threshold and strict  
 112 growth in  $K$ , but not that it is dynamically selected. The dynamical mean-  
 113 field limit—propagation of chaos—is the missing link between the finite sys-  
 114 tem and the stationary description (Sznitman, 1991), and no theorem in this  
 115 development closes it. Exchange symmetry alone does not justify factoriza-  
 116 tion of the joint law, particularly for an ensemble with an unpinned collective  
 117 phase.

118 A spatial version replaces the finite sum by an integral against a kernel

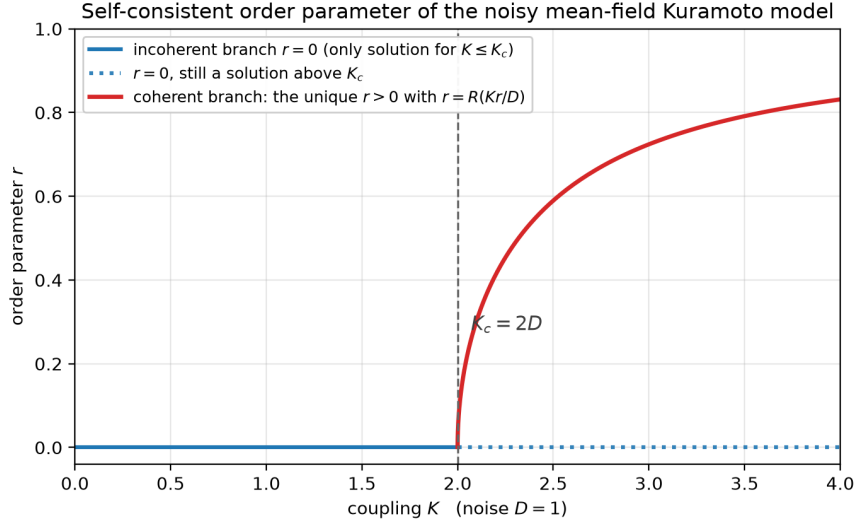


Figure 1: Stationary self-consistency for identical frequencies and phase diffusion  $D = 1$ . The positive coherent branch appears continuously above  $K_c = 2D$  and increases with coupling. Zero remains a solution above threshold. The curve is computed from the same integral expressions used in the formalization; it illustrates the static theorems and does not establish dynamical selection.

119  $K(x, y)$ . We normalize the substrate measure to total mass one, so the mean-  
 120 field strength is a kernel average, not a quantity that grows with the arbitrary  
 121 measure of the substrate. Reduction of a heterogeneous spatial kernel to a  
 122 scalar  $K$  is an additional modelling step, and the threshold in Eq. (2) does  
 123 not automatically apply to every spatial or signed network.

## 124 2.2 Physical identification and calibration

125 Regular mesh refinement gives convergence of a discrete coupling-energy  
 126 quadrature to its continuum integral: scalar energy convergence, not conver-

127 gence of trajectories and not a construction of a physical kernel. The model  
128 posits a continuum kernel and separately identifies its cortical realization  
129 with endogenous electromagnetic coupling. The formal predicate requires  
130 a jointly continuous kernel, a normalized substrate, a mean-field strength  
131 equal to its spatial average, and no single site providing more than half of  
132 that mean-field coupling. Toy witnesses satisfy these conditions; none is a  
133 model calibrated to cortex.

134 Extracellular fields can modulate spike timing (Fröhlich & McCormick,  
135 2010; Anastassiou & Koch, 2015), and field-based models have been proposed  
136 for integrating distributed neural information (Pinotsis & Miller, 2022; Mc-  
137 Fadden, 2020). These observations motivate an effective interaction without  
138 establishing its magnitude relative to phase diffusion, its spatial averaging  
139 rule, or its causal role in experience; changes in phase organization under  
140 propofol (Bastos et al., 2021) likewise do not measure a crossing of the model’s  
141 coupling threshold.

142 The physical-unit conversion is the sharpest open limitation of the frame-  
143 work. An estimate proportional to  $N\Delta tf$ , using population size, a spike-time  
144 shift and oscillation frequency, is dimensionless. It cannot be compared di-  
145 rectly with an inverse-time diffusion parameter. A conversion such as

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

146 requires a rate calibration  $\gamma$ , where  $s$  is the time shift per unit field and  $E$

the field amplitude. Supplemental Material sets out why  $\gamma$  is not recoverable from the other factors: it states how a per-cycle phase advance accumulates into a drift rate, and the choice that would let it cancel removes the very ratio the threshold test is about. The saved field-strength calculations are conditional on this calibration and on the chosen spatial aggregation; they do not independently establish  $K_{\text{eff}} > 2D$  in cortex. Determining the interaction kernel and phase diffusion from the same physical system is necessary before the threshold becomes a quantitative biological claim; Section 3.2 makes that measurement the first item of the proposed experiment.

### 3 Recovery after awakening: a conditional prediction

Sleep-associated changes in extracellular space and clearance motivate a possible slow control of coupling (Xie et al., 2013). We hypothesize that recovery after awakening includes a change in effective coupling  $K(t)$ , mediated by extracellular geometry, on a slower timescale than phase dynamics. Neither the monotonic relation between geometry and coupling nor its role in human sleep inertia follows from the cited physiological observation; the model requires both to be measured.

If  $K(t)$  rises through  $K_c$  and the phase distribution tracks the coherent

166 stationary branch, Eq. (3) predicts

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

167 The ideal stationary description has a subthreshold interval followed by con-  
 168 tinuous onset. Saturation requires a specified subsequent coupling trajectory;  
 169 monotonic coupling alone does not determine a logistic or any unique sig-  
 170 moid. The onset time is fixed by  $K(t_0) = K_c$  only if  $K(t)$  is independently  
 171 known. When coupling is inferred from the same coherence trace, this is not  
 172 a parameter-free prediction of latency.

173 Critical slowing down limits the adiabatic prediction of Eq. (5) near  
 174 threshold, and a finite population carries nonzero fluctuating order below  
 175 it. An observable onset therefore requires a predeclared threshold above the  
 176 finite-size floor, and a finite-rate trajectory need not show an infinite-slope  
 177 cusp. Section 4.1 quantifies the limitation.

### 178 **3.1 Distributional and temporal tests**

179 The stationary model makes a second prediction: phase concentration and  
 180 coherence obey  $r = I_1(a)/I_0(a)$ , with no fitted shape parameter. Its appli-  
 181 cability depends on the stationary-density assumption and the observation  
 182 model, and the relation is common to the whole von Mises family, so agree-  
 183 ment with it identifies neither ephaptic coupling nor this framework against  
 184 alternative synchronization models. Within the same ansatz  $a/r = K/D$  for

185  $r > 0$ , so the limiting threshold ratio follows from the density relation instead  
186 of testing it, and ratios near the finite-size floor are especially unstable.

187 The estimators must be independent, which is the constraint most easily  
188 violated in practice. Estimating  $a$  by inverting the Bessel ratio from  $r$  would  
189 enforce the relation being tested, and linear statistics on wrapped phases  
190 estimate neither quantity.

## 191 **3.2 A protocol for the awakening experiment**

192 The requirements above define a design rather than a single measurement,  
193 and the design is where the content of the hypothesis lives. An adequate test  
194 needs all six of the following.

195 **1. Coupling and diffusion in the same units.** Estimate  $K$  and  $D$   
196 from the same cortical regime, in inverse-time units, separating phase  
197 diffusion from quenched frequency spread and from observation noise.  
198 Without this the threshold  $K > 2D$  is not an evaluable statement  
199 about cortex.

200 **2. Geometry calibrated to coupling.** A diffusion-MRI observable or  
201 an astrocytic signal is an indicator of extracellular geometry, not a mea-  
202 surement of  $K$ . Its mapping to effective coupling must be established  
203 separately, with its own uncertainty.

204 **3. Phase estimation at adequate spatial resolution.** At least 100  
205 spatial samples per estimate. Estimate  $r$  by the circular resultant

length and  $a$  by an unweighted regression of log-density on the cosine of phase relative to the circular mean. With fewer sites, pooling over time changes the estimand to a spatiotemporal distribution and must be stated as such.

4. **An independent recovery endpoint.** Behavioral recovery measured separately from the phase trace, so that latency is not defined by the signal whose latency is at issue.

5. **A predeclared onset criterion.** A coherence threshold fixed above the finite-size floor before unblinding, chosen in the knowledge that a finite ramp smears the cusp.

6. **Model comparison under matched observation.** Coupling-gated recovery compared against exponential, flexible sigmoidal and other slow-recovery models under identical temporal filtering and observation noise. Latency alone cannot distinguish the mechanism, because many processes create delays.

The design fails informatively in three ways: failure of the stationary phase-density relation over a sufficiently broad concentration range; absence of the proposed coupling–geometry association; or a recovery trace better explained by an alternative after finite-rate and observation effects are accounted for. Two apparent failures are not failures. A linear apparent onset is evidence against the adiabatic approximation only where that approximation is independently justified, and it does not falsify the field hypothesis

228 under arbitrary ramp speeds. The absence of a fold in the stationary equa-  
 229 tion excludes stationary coexistence of two positive branches in this model,  
 230 but does not exclude finite-rate hysteresis or adjudicate theories of ignition.

## 231 4 Simulation results

232 The simulations characterize the finite-size, finite-rate regime that the sta-  
 233 tionary theory leaves open, and they bear on whether the prediction of Sec-  
 234 tion 3 is measurable at all. Two of them, Sections 4.1 and 4.3, illustrate  
 235 methodological issues that also arise in analyses of driven transitions and  
 236 dissipation-driven plasticity. Each is a finite run at its sampled parameters  
 237 and not a general statement, and none discharges a hypothesis of the com-  
 238 position in Section 5. Each run uses explicit seeds and compact saved sum-  
 239 maries, and the numerical values below are generated from those artifacts.  
 240 The supplement gives parameters, estimators, failure criteria and additional  
 241 panels. Identical-frequency runs use phase diffusion rather than a quenched  
 242 frequency-spread surrogate, and order plots display the  $1/\sqrt{N}$  reference.

### 243 4.1 Ramp speed sets the apparent onset exponent

244 The dynamic-ramp study (Fig. 2) uses  $N = 2000$ ,  $D = 1$  and 32 replicas at  
 245 four coupling speeds. Escape is the first of three consecutive saved samples at  
 246  $r \geq 0.2$  after threshold. Only 5/32 replicas escape within the fastest window.  
 247 The three slower ramps have mean coupling delays 0.1947, 0.0684 and 0.0253,

248 with a fitted delay exponent 0.443. The apparent onset exponent changes  
249 from 0.964 at the fastest speed to 0.417 at the slowest.

250 The exponent recovered by fitting a driven transition is a joint function  
251 of the mechanism and the drive rate. An onset exponent read off a recovery  
252 trace therefore says little about the underlying transition unless the ramp rate  
253 is independently known and the finite-size floor is declared, and that holds  
254 for any such analysis, not only for one conducted under this framework. The  
255 prediction of Section 3 is the first thing it constrains: the stationary square-  
256 root law cannot be imposed on an arbitrary recovery trace, and its absence  
257 is not by itself evidence against a coupling-gated mechanism. The specified  
258 deviation-based critical ramp speed is not bracketed by the sweep.

259 The dynamical-selection control shows escape toward a coherent branch  
260 above threshold, but retains quantitative finite-size discrepancies. At  $K =$   
261 2.8, steady mean order is 0.67735 against the static value 0.68270. The  
262 fitted growth-rate slope is 0.43240, compared with the theoretical slope  $1/2$ .  
263 The propagation-of-chaos diagnostics show decreasing subcritical dependence  
264 with increasing population size. In the supercritical unpinned ensemble,  
265 however, pair correlation remains substantial because realizations share a  
266 random collective orientation. This negative unconditional result does not  
267 settle a conditional mean-field limit after removal of that orientation.

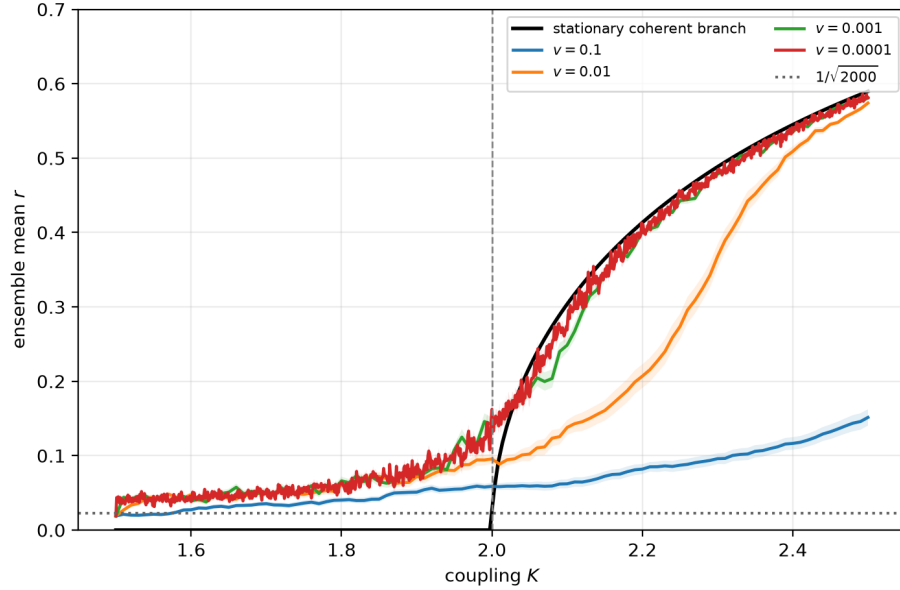


Figure 2: Finite-rate departures from the stationary branch. Ensemble-mean order is plotted against the increasing coupling for four ramp speeds  $v$ , with bands of one standard error across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape; the stationary branch is not a universal time course for recovery.

## 268 4.2 Spatial coupling, frustration and calibration

269 A periodic two-dimensional sheet with row-normalized exponential coupling  
 270 remains coherent in the sampled empirical decay-length band under the de-  
 271 clared total coupling and frequency spread. At decay lengths 0.1, 0.2 and  
 272 0.3 mm, steady order is 0.9424, 0.9427 and 0.9427. This is one finite param-  
 273 eter regime; fixing total coupling while changing range tests fragmentation,  
 274 not whether cortex supplies that total coupling.

275 A row-balanced Dale-law network shows that row balance alone does not

276 guarantee a floor-level incoherent baseline. In a separately specified weak-  
 277 synapse regime, a uniform modulatory term produces an operational coher-  
 278 ence crossing at  $K_{\text{eff}}/D$  between 1.8994 and 1.9387. That ratio is dimension-  
 279 less; carrying it over to cortex needs the rate calibration  $\gamma$  of Section 2.2,  
 280 which neither this sweep nor the spatial control supplies.

### 281 4.3 Squared-drift plasticity does not learn structure

282 A coupled system that descends a dissipation-like objective is widely expected  
 283 to come, in doing so, to mirror the structure of its input; the predictive  
 284 dissipation bound of Section 5 invites that reading. This control tests the  
 285 expectation directly in the present model, and the expectation fails.

286 For fixed phases, consider the squared-drift objective

$$\sigma(K) = \frac{1}{D} \int \left[ \omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

287 It is a quadratic functional of an affine image of the kernel. The formalization  
 288 constructs the bounded drift operator  $A$  and its adjoint on the relevant  $L^2$   
 289 spaces, obtaining  $\nabla\sigma = (2/D)A^*(AK + \omega)$ , and the specified gradient flow  
 290 decreases this objective at fixed phases. The objective is not generally a  
 291 thermodynamic entropy-production rate, which requires probability-current  
 292 information. Whether biological couplings follow this flow is an additional  
 293 modelling question.

294 The joint control evolves noisy phases with clustered frequencies and

295 nonzero grand-mean drive, while updating symmetric couplings every 50 fast  
 296 steps. Couplings remain nonnegative, hollow and normalized to a fixed to-  
 297 tal. The symmetric-edge derivative of  $\frac{1}{2} \sum_i v_i^2$  is  $(v_i - v_j) \sin(\theta_j - \theta_i)$ , so both  
 298 endpoints contribute. Resource constraints and evolving phases remove the  
 299 fixed-phase monotonicity guarantee.

300 Across three seeds, second-half order lies between 0.9811 and 0.9815, and  
 301 matched frozen-coupling controls maintain similar coherence. The distance  
 302 to the true cluster template increases in every adaptive run, with worse  
 303 changes than the shuffled-label comparison (Fig. 3). The template is a co-  
 304 variance of shared latent cluster signals, not an input to the update or a  
 305 measured covariance of the constant frequencies.

306 The objective descends, coherence is maintained, and the coupling matrix  
 307 moves away from the generating structure—below the level of a random rela-  
 308 belling. Coherence is therefore not a proxy for representational alignment in  
 309 this regime, and a plateau in the objective establishes neither stationarity of  
 310 the adaptive process nor a thermodynamic entropy-production rate. What  
 311 fails here is the objective rather than thermodynamically specified learning  
 312 at large: an Ising model whose couplings are annealed degrees of freedom,  
 313 driven by external fields that carry the data, memorizes and generalizes un-  
 314 der ordinary statistical-mechanical rules (Corberi et al., 2026). Those cou-  
 315 plings equilibrate under a Hamiltonian containing the data; these descend a  
 316 squared-drift functional at fixed phases. Which functional the slow variables  
 317 follow is what decides whether they acquire environmental structure.

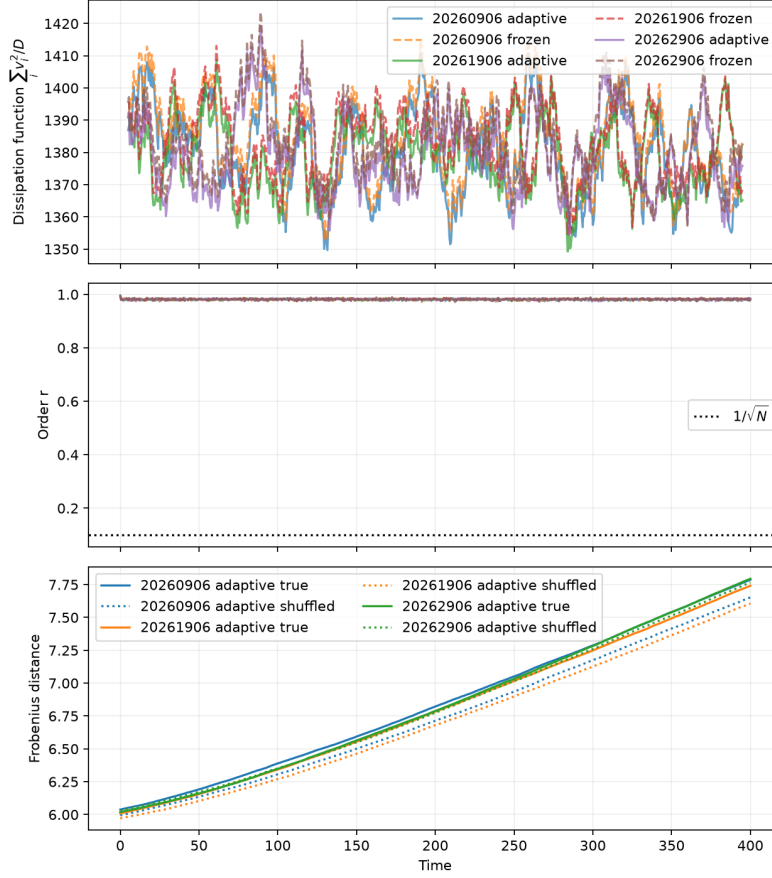


Figure 3: Joint fast/slow dynamics with matched frozen-coupling controls. Top: smoothed squared-drift dissipation, with frozen controls dashed. Middle: order and the finite-size reference. Bottom: distances to true cluster templates (solid) and node-label shuffles (dotted). Coherence persists, but structural distances increase. The result limits the proposed connection between dissipation minimization and environmental structure.

## 318 4.4 The estimation protocol demonstrated on public 319 EEG

320 Item 3 of Section 3.2 specifies estimators whose behavior on real recordings  
321 is not obvious, so we exercise them on public propofol-sedation EEG (Bajwa  
322 et al., 2025). The purpose is to show what the protocol produces and where  
323 it is fragile, not to test the awakening hypothesis, for which these data are  
324 the wrong design.

325 Two fragilities appear immediately. The implemented estimator pools 62  
326 channels over a 100-ms window, so it estimates a spatiotemporal distribu-  
327 tion rather than an instantaneous spatial one—a departure from the protocol  
328 forced by channel count, and one that any scalp implementation will face.  
329 Bipolar and common-reference montages give different residuals, making ref-  
330 erencing a material analysis choice rather than a preprocessing detail.

331 The bipolar results are compatible with the Bessel relation over the ob-  
332 served concentration range. That range is too narrow to distinguish the  
333 curve from its linear approximation at the observed scatter, so the check is  
334 uninformative about the relation itself. An informative test needs a broader  
335 concentration range and explicit treatment of source mixing, finite samples  
336 and temporal pooling; intracranial or high-density recordings across a gen-  
337 uine state transition are the natural setting.

## 338 5 Thermodynamic constraints and the for- 339 mal composition

340 The synchronization model of Section 2 is one link of a longer conditional  
341 construction, whose nine nodes are:

342     **n1** a finite phase space has bounded information capacity; **n2** a  
343     broken symmetry forces a boundary, where the field leaves its vac-  
344     uum manifold; **n3** the boundary erases, and erasure costs heat;  
345     **n4** that dissipation bounds nonpredictive memory; **n5** discrete  
346     couplings coarse-grain to a continuum limit; **n6** a specified phys-  
347     ical field realizes that limiting coupling; **n7** above threshold the  
348     field carries a unique coherent order parameter; **n8** compatible  
349     local states glue to a unique global section; **n9** that section is the  
350     unique fixed point of a self-prediction map.

351 Node-level results are established or cited; what the composition consumes  
352 are the eight implications between them. Figure 4 shows where they sit and  
353 what kind of assumption each one is, and Table 1 states them, classified by  
354 their role in the composition rather than by their plausibility. In particular,  
355 finite capacity does not imply a symmetry-breaking boundary: **n1** and **n2** are  
356 independent roots, even though the composition can accept an implication  
357 between them as an argument.

### 358 5.1 Capacity, boundaries and predictive memory

359 For a finite state space  $X$ , Shannon entropy satisfies  $H(p) \leq \log |X|$ , with  
360 equality only for the uniform distribution. A stream with more possible

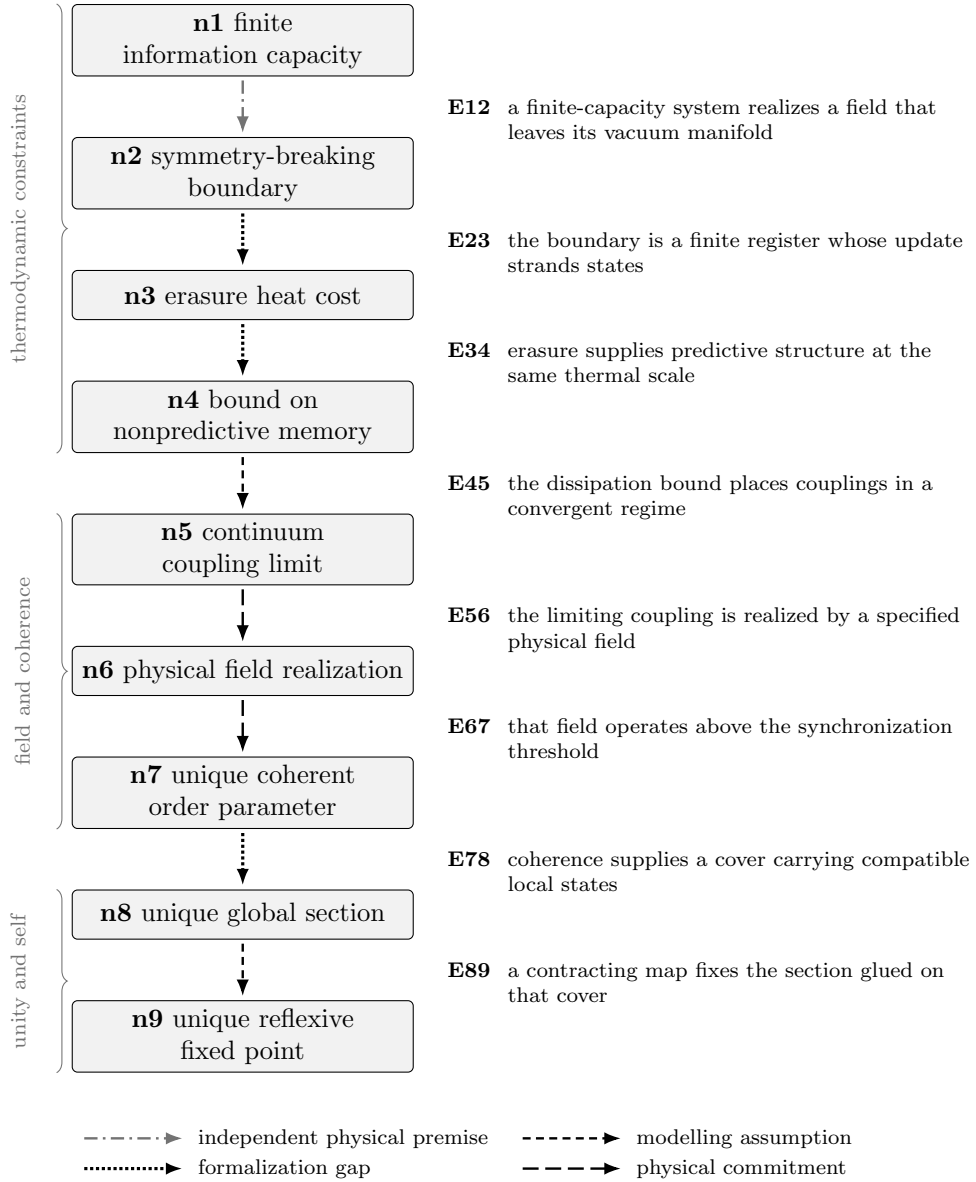


Figure 4: The conditional composition. Each node is an established or cited result; each arrow is an implication the composition consumes and no theorem here supplies, styled by the kind of assumption it carries. The three formalization gaps bracket the two physical commitments, so the chain’s empirical content sits between links of a different kind on either side. E12 is drawn as a premise because n1 and n2 are independent roots. Table 1 states each connection in full.

| Edge                                         | Status                             | Required connection                                                                                                                       |
|----------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| E12<br>capacity $\rightarrow$<br>boundary    | Independent<br>physical<br>premise | A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity alone does not imply this.                            |
| E23<br>boundary $\rightarrow$<br>erasure     | Formalization<br>gap               | The boundary is associated with a finite register whose update leaves states unreachable.                                                 |
| E34<br>erasure $\rightarrow$<br>prediction   | Formalization<br>gap               | Erasures supplies a predictive structure with the same thermal scale and heat budget, and nonzero wasted memory.                          |
| E45<br>prediction $\rightarrow$<br>continuum | Modelling<br>assumption            | A predictive dissipation bound places coupling energies in a convergent coarse-graining regime.                                           |
| E56<br>continuum $\rightarrow$<br>field      | Physical<br>commitment             | The limiting coupling is realized by a specified field satisfying the kernel conditions. Its identification with cortex is empirical.     |
| E67<br>field $\rightarrow$<br>coherence      | Physical<br>commitment             | That field operates above the synchronization threshold.                                                                                  |
| E78<br>coherence $\rightarrow$<br>unity      | Formalization<br>gap               | Coherence supplies a cover with coupling at least the named mean-field value, reached by relaxation and carrying compatible local states. |
| E89<br>unity $\rightarrow$ self              | Modelling<br>assumption            | The state glued on that same cover is fixed by a contracting map mediated by its local self-encoding.                                     |

Table 1: The assumptions consumed by the conditional composition theorem, each an implication between two of the nine nodes above. Node-level results do not establish these connecting implications. The complete theorem map and witnesses are in Supplemental Material, Table S1.

361 histories than  $|X|$  cannot be encoded injectively in that register. Separately,  
 362 a continuous field on a connected substrate that reaches distinct components  
 363 of a disconnected vacuum manifold must leave that manifold somewhere,  
 364 the simplest topological-defect construction (Kibble, 1976). Relating this  
 365 locus to a particular finite information-processing register is the additional  
 366 hypothesis E12 carries.

367 A non-surjective update on a finite register is non-injective. Under the  
 368 specified system–bath model and thermal assumptions, erasure carries a Lan-  
 369 dauer heat cost (Landauer, 1961). Finite capacity does not imply that every  
 370 update erases, and logical many-to-one processing must be distinguished from  
 371 its physical implementation (Norton, 2005; Shenker, 2000). The formaliza-  
 372 tion records the physical heat identity as an obligation of each model and  
 373 supplies an explicit one-bit eraser satisfying it.

374 For a system state  $X_t$  driven by an environmental signal  $S_t$ , the predictive  
 375 dissipation bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (7)$$

376 Under the no-back-action Markov structure  $X_t \rightarrow S_t \rightarrow S_{t+1}$ , data process-  
 377 ing makes the bracket nonnegative (Still et al., 2012). The bound constrains  
 378 nonpredictive memory at a given dissipation budget. It does not specify  
 379 a learning rule, maximize prediction, or require the internal representation  
 380 to reproduce environmental geometry—the point the plasticity control of

381 Section 4 makes concretely. Connecting this budget to Landauer heat ad-  
 382 ditionally requires nonpredictive information to be bounded by the entropy  
 383 erased by the register. The formal bridge derives Eq. (7) from Landauer’s  
 384 bound under that identification; it does not derive the identification.

385 A bound  $D_{KL} \leq \Delta t \sigma$  also cannot force  $D_{KL} \rightarrow 0$  unless its right-hand  
 386 side tends to zero, and Gibbs’ inequality proves nonnegativity of relative en-  
 387 tropy, not a general identification of relative entropy with physical entropy  
 388 production. The chain uses a different technical sense of information at each  
 389 step—a counting bound on a register, a thermodynamic heat cost, a predic-  
 390 tive mutual information—and those senses are not interchangeable (Rosas  
 391 et al., 2026). These distinctions prevent an informational inequality from  
 392 being read as a derivation of environmental learning, Bayesian inference, or  
 393 semantic content.

## 394 5.2 Compatible states and reflexive fixed points

395 A sheaf formalizes local descriptions and their restriction to smaller regions.  
 396 If local sections agree on every overlap, the sheaf condition gives a unique  
 397 global section. In the probability-sheaf construction, this is consistency of  
 398 local density information, and it is not implied by agreement of oscillator  
 399 phases: the development includes a counterexample with equal phases but  
 400 incompatible densities on an overlap. Section 6 rests on this distinction,  
 401 and the model excludes such configurations only through its compatibility  
 402 hypothesis.

403     A thermodynamic cover couples this gluing construction to a phase con-  
 404     figuration at a potential minimum. For the deterministic positive-coupling  
 405     model, convergence to that minimum is proved under an explicit initial basin  
 406     condition; other equilibria prevent an unconditional conclusion. The station-  
 407     ary noisy order parameter, the deterministic relaxation, and the cover must  
 408     therefore be connected by E78 rather than identified without a hypothesis.  
 409     Neither field continuity nor synchronization alone guarantees a globally com-  
 410     patible representation.

411     A reflexive extension equips global sections with a complete metric and a  
 412     predictive map formed by restricting a state to a local encoding region and  
 413     reading that encoding out. If the map contracts and fixes the section supplied  
 414     by the cover, Banach’s theorem gives uniqueness of that same fixed point.  
 415     The contraction rate is assumed, and the equality between the glued section  
 416     and the fixed point is part of E89. A coherent order parameter supplies the  
 417     supercritical inequality required by the stated contraction bound; it does not  
 418     construct the map or prove that cortical dynamics implements it.

419     The theorem `chain` concludes `UnifiedSelf` from these hypotheses. A  
 420     common toy witness satisfies all eight, so their conjunction is not empty.  
 421     The project declares no additional Lean axioms beyond the foundational  
 422     axioms used by Lean’s libraries; physical content resides in hypotheses and  
 423     model fields. This checks the conditional mathematical composition, not the  
 424     empirical truth of its premises. Detailed constructions, scope statements and  
 425     counterexamples are retained in the supplement and source files.

## 426 6 What the construction commits to about 427 consciousness

428 The composition terminates in a mathematical object, and we take a posi-  
429 tion on what that object is. The commitment has two clauses. *Unity*: the  
430 unique global section glued from compatible local states is the formal corre-  
431 late of the unity of a conscious episode—the fact that a subject undergoes  
432 one experience rather than a collection of co-occurring local ones. *Selfhood*:  
433 the fixed point of the self-prediction map on that same section is the formal  
434 correlate of a minimal self, a state that is what it takes itself to be. Neither  
435 clause is a theorem, and no simulation here supports either. We argue for  
436 them, then say what would count against them.

437 Three features of the construction motivate the identification, and each  
438 distinguishes it from a bare synchrony measure.

439 First, compatibility rather than agreement. Oscillator phases can agree  
440 everywhere while the local probabilistic descriptions they carry remain in-  
441 compatible on their overlaps, and Section 5.2 exhibits that configuration  
442 inside the formalization, not only as an illustration. If the unity of experi-  
443 ence is a matter of local contents being about one world rather than merely  
444 co-timed, then compatibility, not coherence, is the property to look for—and  
445 coherence enters as the physical condition under which compatibility can be  
446 maintained, which is what E78 asserts.

447 Second, uniqueness follows from the construction rather than from a sep-

448 arate selection principle. Accounts of conscious unity need something that  
449 picks one experience from among the candidates a system could support;  
450 integrated information theory postulates exclusion for this purpose (Tononi  
451 et al., 2016). Here uniqueness arrives twice as a consequence: given a cover  
452 whose local sections agree, the sheaf condition gives exactly one global sec-  
453 tion, and given a contracting self-prediction map, Banach’s theorem gives  
454 exactly one fixed point. The scope of that claim needs stating exactly, be-  
455 cause the stronger reading is available and is not ours. Uniqueness is relative  
456 to a cover and a map, and choosing them—which regions, which overlaps,  
457 which encoding region, and so where the subject ends—is not supplied by ei-  
458 ther theorem. It is what E78 and E89 carry. What the construction removes  
459 is the need for an additional principle to select among the experiences a fixed  
460 cover and map would otherwise admit; it does not remove the need to say  
461 which cover and map a given subject has. That is a structural argument,  
462 not evidence.

463 Third, the reflexive clause makes the self a condition rather than an addi-  
464 tion. The predictive map restricts a global state to a region encoding it and  
465 reads that encoding back out, so a fixed point is a state whose self-model is  
466 accurate about the state it models. That condition is available only because  
467 the state modelled is a single glued object: a system whose local descrip-  
468 tions do not cohere has no such state for a self-model to be about. Unity  
469 and selfhood are therefore one construction read at two levels, which is why  
470 `UnifiedSelf` requires the fixed point to be the glued section itself rather

471 than an unrelated section.

472     The costs belong with the argument. The commitment does not address  
473 why any physical condition is accompanied by experience; it is a proposal  
474 about which physical state is the conscious one, not about why states of that  
475 kind feel like anything. It is therefore open to the objection that a system  
476 could satisfy every condition—compatible local states, a supercritical field,  
477 an accurate self-model—while nothing is experienced. We have no reply to  
478 that objection and claim none. The identification is also multiply realizable  
479 as stated: nothing in the sheaf or fixed-point conditions is specific to cortex  
480 or to electromagnetic coupling, so the substrate claim is carried entirely by  
481 E56 and E67 and is empirical. No conclusion about artificial systems follows  
482 from the formal results.

483     What could count against the commitment is dissociation. It predicts that  
484 the two clauses and the presence of a unified conscious episode go together,  
485 so a state in which local cortical descriptions are demonstrably compatible  
486 and the field supercritical, while the subject reports no unified experience,  
487 would tell against it; so would unified reported experience where compati-  
488 bility demonstrably fails. Anesthetic emergence, dreaming and dissociative  
489 states are the places to look, because they separate coherence from report  
490 in ways ordinary waking does not. Such tests need an operational measure  
491 of overlap compatibility on neural data, which this article does not supply;  
492 constructing one is the main obstacle to making the commitment empirically  
493 live rather than merely stated. Supplemental Material specifies what such an

observable would have to satisfy—a restriction map, a statistic with a null under decoding noise, validation against configurations whose compatibility is built rather than inferred—and the three ways it would fail silently, one of which is reading compatibility off synchrony and so assuming away the distinction being tested.

The physical tests of Sections 3 and 4 are independent of all of this. They concern coupling, coherence and recovery, and can fail while the identification stands, or succeed while it is wrong.

## 7 Discussion

The framework connects thermodynamic information constraints, collective phase organization and compatible local representations, conditionally and with its assumptions named. Its biological content is a coupling-gated recovery hypothesis with defined observables, a stated protocol, and failure conditions declared in advance. Its formal content is a composition whose eight gaps are exhibited rather than argued past, and which can be inhabited by a single model. Its interpretive content is the identification of Section 6, offered as a commitment with reasons.

The simulations are useful because they restrict the hypothesis rather than support it. Finite ramps smear the stationary onset to the point where a fitted onset exponent chiefly reports the ramp; an unpinned collective phase makes independence diagnostics ambiguous in exactly the supercritical

515 regime the model cares about; and squared-drift plasticity maintains coher-  
516 ence while degrading structural alignment. The second and third of these  
517 are results about method and about a common intuition, and they transfer  
518 beyond this framework.

519 The proposal differs in emphasis from accounts based on integrated in-  
520 formation (Tononi et al., 2016), global workspace dynamics (Baars, 1997;  
521 Dehaene et al., 2014), or predictive processing (Friston, 2010). It specifies a  
522 candidate interaction medium and a mathematical compatibility condition,  
523 and, relative to a given cover and self-prediction map, it derives the unique-  
524 ness those accounts postulate. It does not derive the exclusion principle of  
525 integrated information, identify every broadcast with a field transition, or  
526 derive a generative model from a dissipation bound. Continuous fields can  
527 carry incompatible local descriptions and discrete models can admit sheaf  
528 constructions, so continuity alone does not settle the combination problem;  
529 compatibility is what this account offers in its place.

530 Two formal results bear on the substrate question, and both admit a  
531 stronger reading than they support. At matched coupling resource, a fixed  
532 wiring support that excludes a better-correlated pair can be suboptimal—a  
533 statement about constrained optimization, not about biology. On an atom-  
534 less substrate, a kernel supported on finitely many points contributes zero to  
535 the chosen continuum coupling integral. That is a property of the measure  
536 and the observable, and it cuts against this framework’s own coarse-graining  
537 step as much as against anything else. It does not imply that digital hard-

538 ware cannot approximate a field or support consciousness, and no biological-  
539 versus-artificial conclusion follows.

540 The empirical priorities follow directly from Section 3.2: estimate effec-  
541 tive coupling and diffusion in consistent units, test the proposed relation  
542 between coupling and extracellular geometry, and measure recovery against  
543 an independently defined behavioral endpoint under matched model compar-  
544 ison. A favorable concentration–coherence relation would support a statis-  
545 tical description, not identify its physical cause; an unfavorable result in a  
546 regime satisfying the stated assumptions would constrain that description.  
547 Successful synchronization simulations cannot substitute for measurements  
548 distinguishing ephaptic interactions from other coupling mechanisms.

549 The article therefore asks to be judged in three separable parts: a pre-  
550 diction that can be tested, a composition that can be checked, and an iden-  
551 tification that can be argued about. Keeping them separate is what makes  
552 the whole assessable.

## 553 **Data and code availability**

554 The Lean development, simulation scripts, tests and saved numerical sum-  
555 maries are maintained in the manuscript source repository. Publication sim-  
556 ulation values are regenerated from saved summaries without rerunning pro-  
557 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset  
558 ds005620 (Bajwa et al., 2025); preprocessing and selection are described

559 in Supplemental Material. The source repository contains the reproducible  
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## 564 **Use of AI tools**

565 AI assistance to this work was substantial and is not separable by task.  
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568 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code  
569 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude  
570 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them  
571 they contributed to the Lean development, the simulation and analysis code,  
572 and the drafting and revision of both documents. Because the tools were  
573 used in combination and in alternation, no section, proof, figure or result is  
574 attributable to any one of them, and none is claimed to be. The author is  
575 responsible for all content, including everything a tool produced. Machine  
576 checking establishes the stated Lean results under their hypotheses; it does  
577 not validate biological interpretations or replace scientific review of generated

578 code and prose.

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